

Distinctive Whole-brain Cell-Types Strongly Predict Tissue Damage Patterns in Eleven Neurodegenerative Disorders

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Abstract

For over a century, brain research narrative has mainly centered on neuron cells. Accordingly, most whole-brain neurodegenerative studies focus on neuronal dysfunction and their selective vulnerability, while we lack comprehensive analyses of other major cell-types' contribution. By unifying spatial gene expression, structural MRI, and cell deconvolution, here we describe how the human brain distribution of canonical cell-types extensively predicts tissue damage in eleven neurodegenerative disorders, including early- and late-onset Alzheimer's disease, Parkinson's disease, dementia with Lewy bodies, amyotrophic lateral sclerosis, frontotemporal dementia, and tauopathies. We reconstructed comprehensive whole-brain reference maps of cellular abundance for six major cell-types and identified characteristic axes of spatial overlapping with atrophy. Our results support the strong mediating role of non-neuronal cells, primarily microglia and astrocytes, on spatial vulnerability to tissue loss in neurodegeneration, with distinct and shared across-disorders pathomechanisms. These observations provide critical insights into the multicellular pathophysiology underlying spatiotemporal advance in neurodegeneration. Notably, they also emphasize the need to exceed the current neuro-centric view of brain diseases, supporting the imperative for cell-specific therapeutic targets in neurodegeneration.

One Sentence Summary

Major cell-types distinctively associate with spatial vulnerability to tissue loss in eleven neurodegenerative disorders.

eLife assessment

Pak et al. examined the relationship between the most common spatial patterns of neurodegeneration and the density of different cell types in the cerebral cortex. The study uses innovative methods but the main claims are **incompletely** supported due to some limitations. While the results might be considered preliminary, this work provides **valuable** findings and takes a step in the right direction by highlighting the contribution of non-neuronal cell type to the pathobiology of neurodegeneration.

Introduction

Neurodegenerative diseases (NDs) are characterised by substantial neuronal loss in the central and peripheral nervous system¹. In dementia-related conditions like Alzheimer's disease (AD), frontotemporal dementia (FTD), and dementia with Lewy bodies (DLB), neurodegeneration can lead to progressive damage in brain regions related with memory, behaviour, and cognition². Other NDs are thought to primarily affect the locomotor system, including motor neurons in amyotrophic lateral sclerosis (ALS) and nigrostriatal dopaminergic circuitry in Parkinson's disease (PD)³. Although each disorder has its own distinct etiology, progression, affected brain areas, and clinical manifestations, transcriptomics analyses support that most of them share molecular and cellular mechanisms⁴⁻⁷.

While research has been mainly focused on neuronal dysfunction, supporting cells such as astrocytes, microglia, oligodendrocytes, cells of the vascular and peripheral immune systems and their contribution to the disease pathology are gaining more recognition⁸⁻¹⁰. Depending on the disease stage, non-neuronal cells in the brain can play a dual role, both protective and detrimental, by producing neuroprotective and pro-inflammatory factors. Thus, their complex response can have both positive and negative effects on neuronal health and survival. For example, the activation of glial cells can lead to metabolic stress, disruption of the blood-brain barrier, and reduced energy, all of which contribute to increased neuronal death¹¹⁻¹⁵. In AD, for example, overproduction of pro-inflammatory cytokines leads to the accumulation of amyloid- β and tau plaques, which are known as the major hallmarks of the disease^{12, 16, 17}. This process depends on the innate immune system, which includes microglia and astrocytes^{18, 19}. Peripheral interference from systemic inflammation or the gut microbiome can also alter the progression of neuronal loss²⁰. Growing evidence suggests that these immune-mediated events is a driving force behind the wide range of NDs^{18, 19, 21, 22}. Yet, the exact bases behind how these processes contribute to selective neuronal loss across brain regions remain unclear.

Recent studies have suggested that brain spatial patterns in gene expression are associated with regional vulnerability to some neurodegenerative disorders and their corresponding tissue atrophy distributions²³⁻²⁶. Comparison of transcriptomic patterns in middle temporal gyrus across various brain diseases showed cell-type expression signature unique for neurodegenerative diseases⁷. Although single-cell transcriptomics and proteomics analyses have advanced our knowledge of cell-type compositions associated with pathology in neurodegeneration²⁷⁻²⁹, these are invariably restricted to a few isolated brain regions, usually needing to be preselected at hand for each specific disease. Due to the invasive nature of tissue acquisition/mapping and further technical limitations for covering extended areas³⁰, no whole-brain maps for the abundance of cell populations in humans are currently available, constraining the analysis of large-scale cellular vulnerabilities in neurological diseases. Accordingly, how spatial cell-types distributions relate to stereotypic regional damages in neurodegeneration remain largely unclear³¹.

Here, we extend previous analysis of cellular-based spatiotemporal vulnerability in neurodegeneration in three fundamental ways. First, we use transcriptomics, structural magnetic resonance imaging (MRI), and advanced cell deconvolution to construct whole-brain reference maps of cellular abundance for six canonical cell-types: neurons, astrocytes, oligodendrocytes, microglia, endothelial cells, and oligodendrocyte precursors. Second, we describe the spatial associations of each canonical cell-type with atrophy maps from eleven low-to-high prevalent neurodegenerative conditions, including early- and late-AD, PD, FTD, DLB, ALS, tau, and TAR DNA-binding protein 43 (TDP43) pathologies. Third, we identify distinctive cell-cell and disease-disease axes of spatial susceptibility in neurodegeneration, obtaining new insights about across-diseases (dis)similarities in underlying pathological cellular systems. We confirm that non-neuronal cells express substantial vulnerability to tissue loss and spatial brain alterations in most studied neurodegenerative conditions, with distinct and shared across-cells and across-diseases mechanisms. This study aids in unraveling the commonalities across a myriad of dissimilar neurological diseases, while also revealing cell-type specific patterns conferring increased vulnerability or resilience to each examined disorder. For further translation and validation of our findings, all resulting analytic tools and cells abundance maps are shared with the scientific and clinical community.

Results

Multimodal data origin and unification approach

We obtained whole-brain voxel-wise atrophy maps for eleven neurodegenerative conditions, including early- and late-onset Alzheimer's disease (EAD and LOAD, respectively), Parkinson's disease (PD), amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), dementia with Lewy bodies (DLB) and dementia-related pathologies (presenilin-1, TDP43A, TDP43C, 3RTau, and 4RTau; see *Materials and Methods, Disease-specific atrophy maps* subsection)^{32, 33}. Pathological diagnosis confirmation was performed for AD and related dementias (DLB, presenilin-1, TDP43A, TDP43C, 3RTau, and 4RTau; ³²), while PD, ALS, and FTD were diagnosed based on clinical and/or neuroimaging criteria^{34, 35}, with some ALS patients being histologically confirmed *post-mortem*³⁵. Change in tissue density in the atrophy maps was previously measured by voxel- and deformation-based morphometry (VBM and DBM; *Materials and Methods, Disease-specific atrophy maps* subsection) applied to structural T1-weighted MR images, and expressed as a t-score per voxel (relatively low negative values indicate greater GM tissue loss/atrophy; ^{36, 37}). All maps are registered to the Montreal Neurological Institute (MNI) brain space³⁸. In addition, we obtained bulk transcriptomic data for the adult human brain from the Allen Human Brain Atlas (AHBA)³⁹. This included high-resolution coverage of nearly the entire brain, measuring expression levels for over 20,000 genes from 3702 distinct tissue samples of six post-mortem specimens, and detailed structural MRI data (see *Materials and Methods, Mapping gene expression data*)³⁹.

Using previously validated approaches to infer gene expression levels (in AHBA data) at not-sampled brain locations⁴⁰, mRNA expression levels were completed for all gray matter (GM) voxels in the standardized MNI brain space³⁸. Next, at each GM location, densities for multiple canonical cell-types were estimated using the Brain Cell-type Specific Gene Expression Analysis software (BRETIGEA)⁴¹. This deconvolution method⁴¹ accurately estimates cell proportions from bulk mRNA for six major cell-types (**Fig. 1C**): neurons, astrocytes, oligodendrocytes, microglia, endothelial cells, and oligodendrocyte precursor cells (OPCs). Overall, atrophy levels for eleven neurodegenerative disorders and proportion values for six major cell-types were unified at matched and standardized locations (MNI space) covering the brain's whole gray matter (see **Fig. 1** for schematic description).

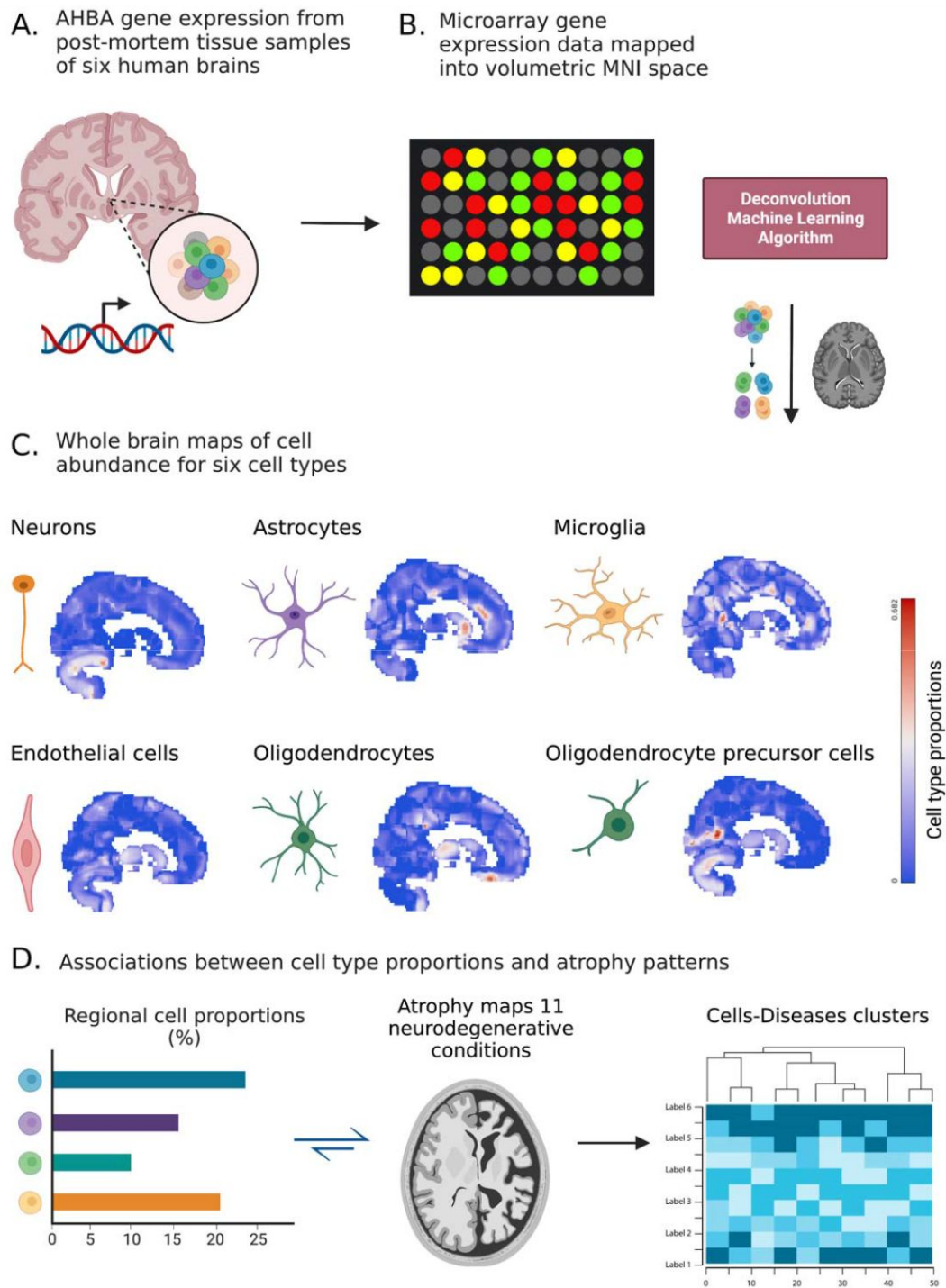


Figure 1.

Schematic approach for whole-brain cells proportions vulnerability analysis in neurodegeneration. **(A)** Gene expression levels in the AHBA were derived from 3072 distinct tissue samples of six post-mortem human brains. **(B)** Microarray gene expression data was inferred for each unsampled GM voxel and together with AHBA data was mapped into volumetric MNI space resulting in the brain-wide transcriptional atlas. Deconvolution machine learning algorithm for bulk RNA expression levels was applied to the transcriptional atlas with using well-known cell-type-specific gene markers. **(C)** Comprehensive volumetric maps showing reconstructed distributions of six canonical cell-types, including neurons, astrocytes, microglia, endothelial cells, oligodendrocytes and OPCs, across all gray matter voxels in the brain (see *Materials and Methods, Cell-type proportion estimation* subsection). At each voxel, red and blue colors indicate high and low proportion densities, respectively. **(D)** Associations between cell-types proportions from each density map and atrophy values in eleven neurodegenerative conditions were analysed in 118 predefined by the AAL atlas regions. Created with BioRender.com (<http://biorender.com/>)

We hypothesized (and tested in next subsections) that brain tissue damages in neurodegenerative disorders are associated with distinctive patterns of cells distributions, with alterations on major cell-types playing a key role on the development of each disorder and representing a direct factor contributing to brain dysfunction.

Uncovering spatial associations between cell-type abundances and tissue damage in neurodegeneration

First, we investigated whether stereotypic brain atrophy patterns in neurodegenerative disorders have systematic associations with the spatial distribution of canonical cell-type populations. For each disease and cell-type pair, the non-linear Spearman correlation was calculated with paired atrophy-cell proportion values across 118 cortical and subcortical regions defined by the automated anatomical labelling (AAL) atlas (Table S1; [42](#)). The results (**Figs. 2A-L**) show clear associations for all the studied diseases, suggesting extensive cell-types related tissue damage vulnerability in NDs. We confirmed that the observed relationships are independent of brain parcellation, obtaining equivalent results for a different brain parcellation (i.e., DKT atlas [43](#); see Fig. S1).

As shown in **Figs. 2A-L**, astrocytes and microglia cell occurrences presented the strongest spatial associations with atrophy in most neurodegenerative conditions, particularly for EOAD, LOAD, DLB, presenilin-1, 3RTau, 4RTau, TDP43A, TDP43C and FTD (all $P < 0.001$). Astrocytes are involved in neuronal support, extracellular homeostasis, and inflammatory regulation in response to injury, and show high susceptibility to senescence and oxidative damage^{[44](#), [45](#)}. Astrocytes also play an important role in the maintenance of the blood-brain barrier (BBB)^{[46](#)}. Atrophy patterns in cortical regions of genetic FTD was shown to be associated with higher expression of astrocytes- and endothelial cells-related genes^{[26](#)}. There is evidence of endothelium degeneration and vascular dysfunction in AD and PD^{[47](#), [48](#)}. This degeneration may lead to disruption of the BBB, which would allow harmful substances to enter the brain, including inflammatory molecules and toxic aggregated proteins^{[48](#), [49](#)}. Furthermore, the endothelium plays a key role in delivering oxygen and nutrients. The disruption of this process in neurodegeneration may lead to impaired brain function and cognitive decline^{[14](#), [50](#)}.

Similarly involved in neuronal support, microglial cells are the resident macrophages of the central nervous system and key players in the pathology of neurodegenerative conditions including EOAD, LOAD, PD, FTD and ALS^{[12](#), [51](#), [52](#)}. Besides its many critical specializations, microglial activation involvement in prolonged neuroinflammation is of particular relevance in NDs^{[11](#), [53](#)}. At earlier stages of AD, increased population of microglia and astrocytes (microgliosis and astrogliosis) have been observed in diseased regions, due to sustained cellular proliferation in response to disturbances, loss of homeostasis or the accumulation of $A\beta$ ^{[13](#), [54](#), [55](#)}. Excessive proliferation leads to the transition of homeostatic microglia to its senescent or disease-associated type, also called DAM in mice models, via the processes mediated by *TREM2-APOE* signalling^{[54](#), [56](#), [57](#)}. Increased number of dystrophic microglia, a form of cellular senescence characterized as beading and fragmentation of the branches of microglia, has been seen in multiple NDs such as AD, DLB and TDP-43 encephalopathy^{[58](#)}. The presence of senescent microglia is believed to ultimately contribute to the failure of brain homeostasis and to clinical symptomatology^{[57](#), [59](#), [60](#)}.

Other non-neuronal cells such as oligodendrocytes and endothelial cells also associated with spatial tissue vulnerability to all NDs aside ALS (**Figs. 2J, L**). Oligodendrocytes are responsible for the synthesis and maintenance of myelin in the brain^{[61](#)}. Demyelination produces loss of axonal insulation leading to sensory or motor neuron death in AD and ALS^{[61](#), [62](#)}. Oligodendrocytes were shown to be highly genetically associated with PD^{[63](#)-[65](#)} and be particularly vulnerable to the alpha-synuclein accumulation^{[66](#), [67](#)}. Mature oligodendrocytes may be in direct contact with the vascular basement membrane, allowing the direct signaling and

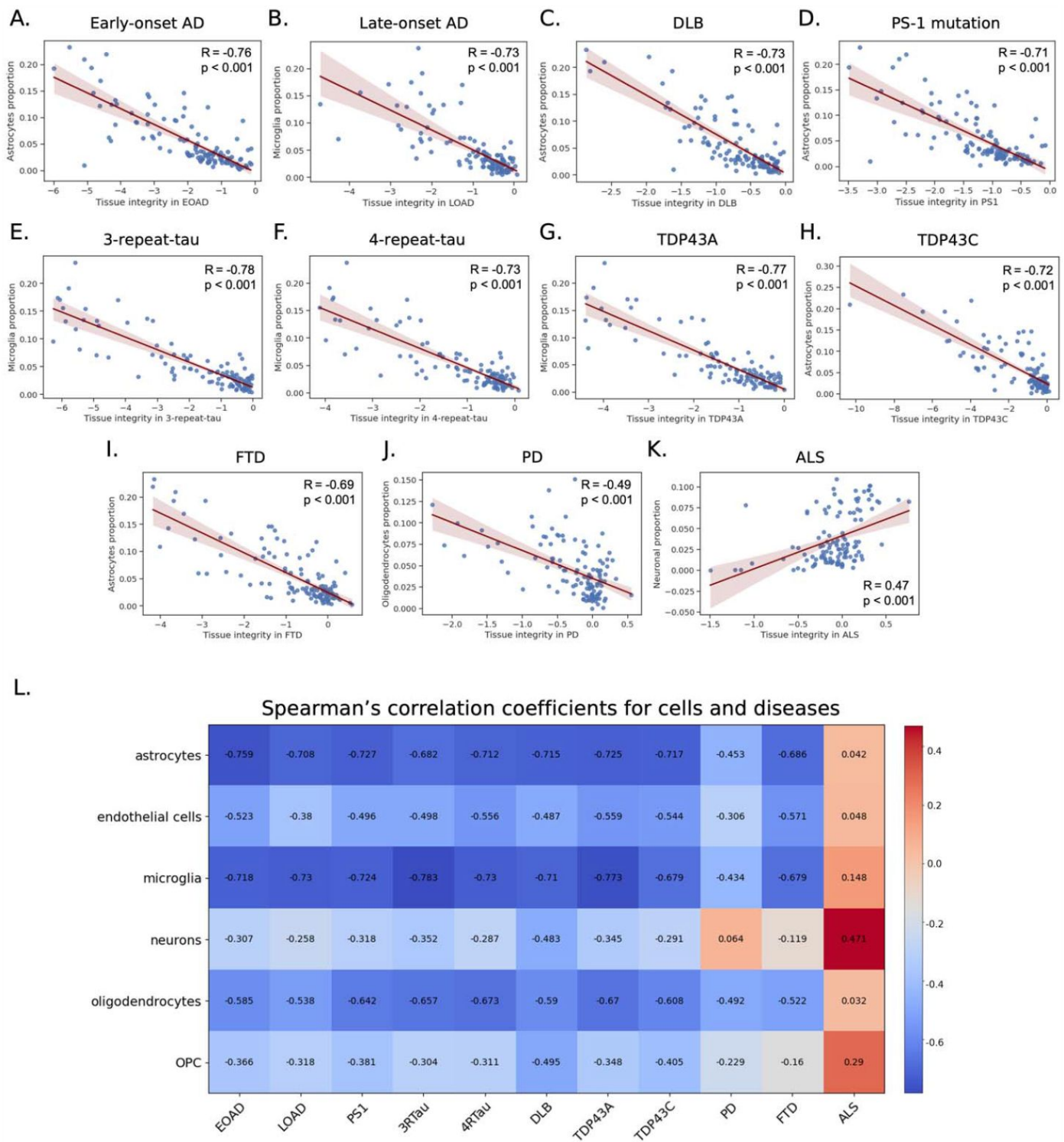


Figure 2.

Spatial associations between tissue integrity and cell-types proportions for eleven neurodegenerative conditions. A-K) Strongest Spearman correlations for EOAD, LOAD, DLB, PS1, 3RTau, 4RTau, TDP43A, TDP43C, FTD, PD, and ALS, respectively. L) All pairwise cells-diseases correlation values. In A-K, each dot represents a GM region from the AAL atlas (Table S1; see Fig. S1 for equivalent results for the DKT parcellation). Lower tissue integrity score indicates greater GM loss/atrophy. Notice how astrocyte density significantly correlates with increase in tissue loss in EOAD, DLB, PS1, TDP43C, and FTD (A, C, D, H, I; $P < 0.001$). Tissue loss was also associated with increase in microglial proportion in LOAD, 3RTau, 4RTau, and TDP43A (B, E, F, G; $P < 0.001$), and increase in oligodendrocytes in PD (J; $p < 0.001$). Increase in neuronal proportion showed association with decrease in atrophy and tissue enrichment in ALS (K; $p < 0.001$).

exchange of substances with endothelial cells^{68, 69}. This particular location could underlie specific functions of vasculature-associated oligodendrocytes, which may contribute to the observed oligodendrocytes-mediated tissue vulnerability in NDs⁶⁸. In addition, densities of OPCs showed strong correlations with the atrophy patterns of DLB and TDP43C. OPCs regulate neural activity and harbor immune-related and vascular-related functions⁷⁰. In response to oligodendrocyte damage, OPCs initiate their proliferation and differentiation for the purpose of repairing damaged myelin⁷¹. In AD, PD and ALS, the OPCs become unable to differentiate and their numbers decrease, leading to a reduction in myelin production and subsequent neural damage^{72, 73}. Curiously, depletion of TDP-43 proteins (in TDP-43A&C, FTD and ALS) may provoke the progressive loss of oligodendrocytes and OPCs^{74, 75}.

We observed (**Fig. 2**) that neuronal abundance distribution is also associated with tissue damage in many neurodegenerative conditions. However, these associations are less strong than for other cell-types, except for the ALS case (**Fig. 2K**). For this disorder, neuron proportions positively correlated with tissue integrity (i.e., the higher the neuronal proportion the less atrophy in a region). This observation suggests that increased neuronal presence at brain regions (relative to all considered cell-types) may have a protective effect in ALS, making neuronal enriched regions less vulnerable to damage in this disorder. In addition, we observed particularly weak associations between neuronal proportions and tissue damage in FTD and PD (**Fig. 2L**), suggesting that these disorders may be primary associated with supportive cell-types (astrocytes and oligodendrocytes, respectively; **Figs. 2I, J**).

Spatial cell-types grouping exposes distinctive disease-disease mechanisms

Next, we hypothesized that diseases sharing similar biological mechanisms and clinical manifestations present common across-brain patterns of cell-types density associations. **Figure 3** shows a hierarchical taxonomy dendrogram grouping cell-types and diseases according to their common brain-wide correlation patterns. We observed that AD-related conditions formed its own cluster with other tau pathologies. Also, patterns of cell-types vs tissue loss in both PD and ALS (known for causing major motor symptoms) were distinct compared with the group of dementias, which primary affect cognitive functioning.

Patterns in cellular vulnerability in DLB did not strongly resemble PD without dementia (**Fig. 3C**), although both conditions involve alpha-synuclein aggregates⁷⁶. This observation supports the hypothesis that clinical deterioration in DLB and PD is not uniquely caused by alpha-synuclein neurodegeneration alone but by distinctive regional spreading patterns^{77, 78}. Previously, differences in cell-type-specific gene expression and bulk-tissue samples differential splicing distinguished PD from the Lewy body dementias, with PDD and DLB demonstrating more similarities⁶⁴. Similar observation can be made for ALS and FTD, despite the presence of TDP-43 accumulations in both conditions⁷⁹ and their strong genetical overlap⁸⁰, they did not group together and with FTLD-tau pathologies (3RTau, 4RTau, TDP43A) based on patterns of cellular vulnerabilities. These results emphasize the fundamental role of network topology (beyond the presence of toxic misfolded proteins) in developing characteristic tissue loss and clinical manifestations in NDs^{33, 81-83}.

Among all cell-types, neurons and OPCs spatial density distributions were least associated with tissue atrophy in all eleven disorders, subsequently clustering together. Astrocytes and microglia distributions showed the strongest associations with dementia-related pathologies, and thus formed a separate group while still being related with oligodendrocytes and endothelial cells. Astrocytes and microglia are known to be intimately related in the pathophysiological processes of NDs¹³. Both are key regulators of inflammatory responses in the central nervous system, and given their role in clearing misfolded proteins, dysfunctions of each of them can result in the accumulation of A β and tau^{13, 17}. During the progression of AD and PD, resident microglia

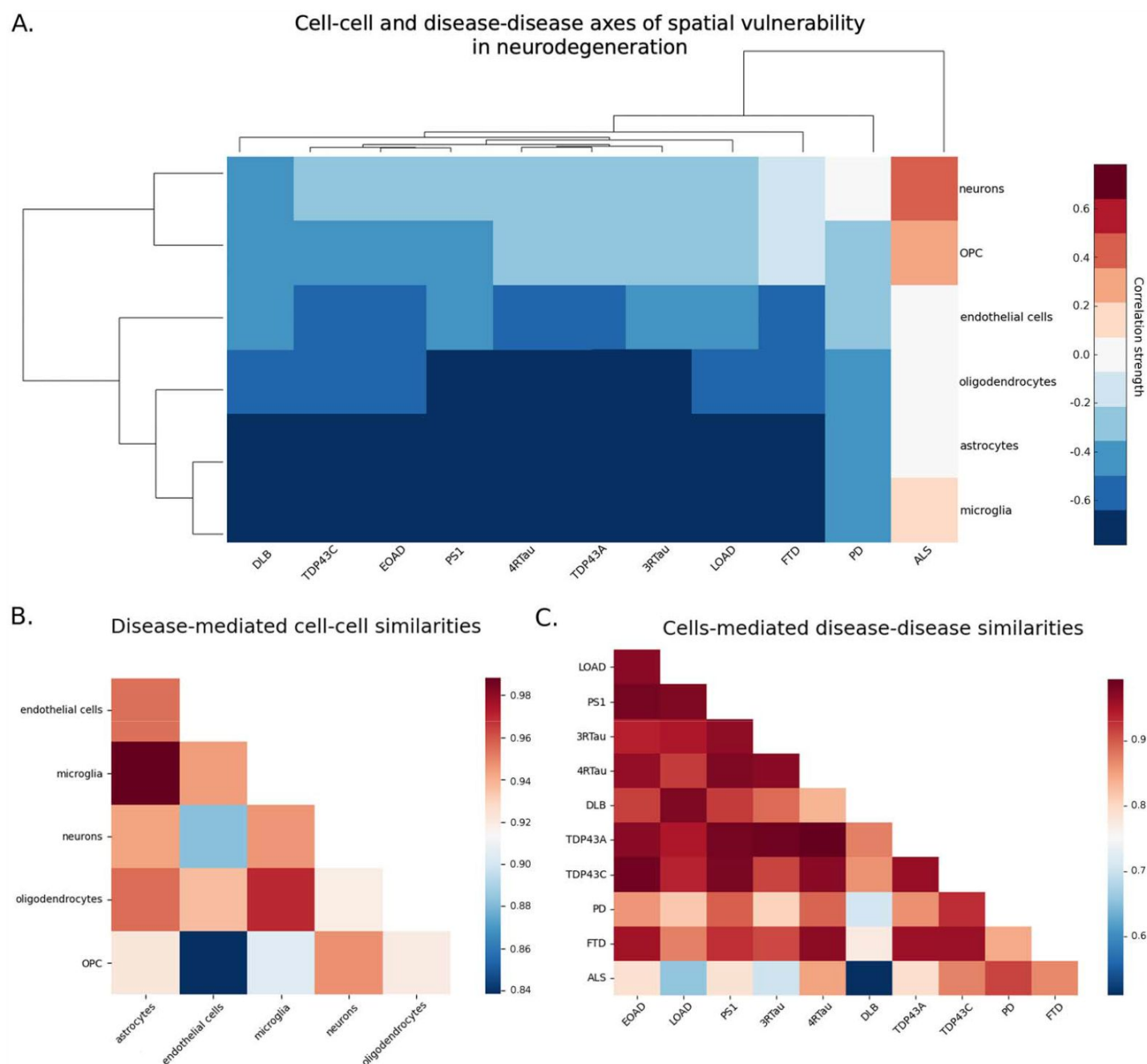


Figure 3.

Cells and diseases similarities based on shared distributions. A) Dendrogram and unsupervised hierarchical clustering heatmap of Spearman's correlations between cell-type proportions and atrophy patterns of the eleven neurodegenerative diseases. B) Cell-cell associations based on regional vulnerabilities to tissue loss across neurodegenerative conditions. C) Disease-disease similarities across cell-types. In A), red color corresponds to strong positive correlations between cells and diseases, white to no correlation, and dark blue to strong negative correlation.

undergo proinflammatory activation, resulting in an increased capacity to convert resting astrocytes to reactive astrocytes⁸⁴. Recent studies demonstrated that blocking the microglial-mediated conversion of astrocytes to a neurotoxic reactive phenotype could limit neurodegeneration in AD and PD^{84, 85}.

Discussion

Previous efforts to describe the composition of the brain's different cell populations related to neurodegeneration have been limited to a few isolated regions. In the most systematic study of its kind, here we characterized large-scale spatial associations between canonical cell-types and brain tissue loss across all cortical and subcortical gray matter areas in eleven neurodegenerative disorders (including early- and late-AD, PD, FTD, DLB, ALS, tauopathies). Starting from gene expression and structural MRIs from the Allen Human Brain Atlas³⁹, and extending our analysis with advanced RNAseq cells deconvolution approaches and whole-brain atrophy maps from clinically and/or neuropathologically confirmed diseases. We determined that (i) the spatial distributions of non-neuronal cell-types, primarily microglia and astrocytes, are strongly associated with the spread tissue damage present in many neurodegenerative disorders; (ii) cells and diseases define major axes that underlie spatial vulnerability according with shared cellular mechanisms, which aid in comprehending heterogeneity behind distinct and similar clinical manifestations/definitions; (iii) the generated whole-brain maps of cellular abundance can be similarly used for studying cellular processes in other neurological conditions (e.g., neurodevelopmental and neuropsychiatric disorders). Overall, our findings stress the critical need to surpass the current neuro-centric view of brain diseases and the imperative for identifying cell-specific therapeutic targets in neurodegeneration⁵⁹. For further translation and validation, all resulting cells abundance maps and analytic tools are freely shared with the community.

We derived, first to our knowledge, high resolution maps of cellular abundance/proportion in the adult human brain for six canonical cell-types, including astrocytes, neurons, oligodendrocytes, microglia, and endothelial cells. As mentioned, previous cellular analyses of neurological conditions have been restricted to expert-selected isolated brain areas. The invasive nature of expression assays, requiring direct access to neural tissue, and other numerous scaling limitations have impeded extensive spatial analyses⁸⁶. Earlier studies, also using AHBA data, have shown that spatial patterns in gene expression and cell-type-specific genetic markers are associated with both regional vulnerability to neurodegeneration and patterns of atrophy across the brain^{7, 23, 24, 26}. Since many neurodegeneration-related genes have similar levels of expression in both affected and unaffected brain areas⁸⁷, characterizing changes in tissue loss associated with cell-type proportions may provide a clearer perspective on large-scale spatial patterns of cellular vulnerability. Our maps of cells-abundance are available for the scientific and clinical community, potentially allowing researchers to further study spatial variations in cell-types density with macroscale phenotypes. These maps can be used in future studies concerning brain structure and function in both health and disease. They can be also explored in context of other neurological diseases including neurodevelopmental and psychiatric conditions.

Our results demonstrate that all canonical cell-types express vulnerability to dementia-related atrophy of brain tissue, suggesting the disruption of the molecular pathways involving specific cell-types can contribute to their observed dysfunctions and subsequent clinical symptomatology⁵⁹. Previously, transcriptional profiling of prefrontal cortex in AD showed reduced proportions of neurons, astrocytes, oligodendrocytes, and homeostatic microglia⁸⁸. In contrast, bulk-RNA analysis of diseased AD tissues from various human brain regions observed neuronal loss and increased cell abundance of microglia, astrocytes, and oligodendrocytes^{89, 90}. Furthermore, increased microglial, endothelial cells, and oligodendrocytes population was observed in PD and other Lewy diseases^{64, 91}. Cortical regions exhibiting the most severe atrophy in FTD showed increased gene expression of astrocytes and endothelial cells²⁶. In line

with these results, we observed that regions with increased cell-type proportions, particularly for astrocytes and microglia, are more vulnerable to tissue atrophy in almost all neurodegenerative conditions. This may partly explain the reported cellular proliferation through microglial activation in diseased regions in response to the misfolded protein accumulation or other pathobiological processes^{54, 60}. As disease progresses, the release of inflammatory agents by sustained microglial activation is believed to be responsible for exacerbating neurodegeneration and clinical symptoms^{18, 21}. Microglial activation in pair with gray matter atrophy in frontal cortex were shown to be directly associated with cognitive decline in FTD⁵².

Taken together, our findings support genome-wide association and transcriptional studies suggesting that cell-type-specific alterations in NDs may indicate the disruption of the following distinct and overlapping molecular pathways: inflammation, immune response, and homeostasis (astrocytes and microglia), vascular system (endothelial cells), and myelination (oligodendrocytes and OPCs)^{88, 92, 94}. Recent study identified a degree of similarities (grouping) in cell-type expression between AD, FTD, and ALS, distinguishing them from other neurological disorders⁷. Interestingly, our results showed all dementia conditions were similar in their patterns of cell-types associations with tissue loss relative to PD and ALS. AD-associated changes in cell-types proportions were previously suggested to be related to the presence of dementia, not to plaques and tangles⁹⁵. There were no strong similarities in patterns of vulnerabilities between pairs of conditions (PD without dementia and DLB, FTD and ALS) with associated genetic risks and type of toxic proteins^{76, 79, 80}. These results reinforce our statement on gray matter atrophy patterns in NDs being not uniquely provoked by genes and misfolded proteins toxicity alone but affected and mediated by additional factors such as metabolic and vascular dysregulations^{81, 96, 98}.

Our study also has a number of limitations. First, our analyses were focused on stereotypic atrophy patterns for each disorder. It is known that NDs are highly heterogenous, with molecular, phenotypic, and clinical subtypes potentially varying in atrophy patterns^{99, 100}. Further investigation of cell-type signatures across various subtypes and disease stages may better characterize each case. Similarly, it has been observed that cell-type-related transcriptional changes are different between sexes⁷⁰, making future sex-specific analyses indispensable for further understanding of sex-related pathomechanisms. Examined atrophy maps were coming from different studies (Table S2), with differences in data acquisition protocols (e.g., spatial resolution) and technical procedures (e.g., smoothing level, statistical methods). In complementary analyses, we observed almost identical results after smoothing all disease-specific images with the same kernel size, while they were already mapped at the same spatial resolution for this study and statistically adjusted by acquisition parameters (e.g., field strength) in original studies. Cell-types deconvolution approaches are varied and limited in their precision¹⁰¹. Here, we used a previously validated deconvolution method designed for efficiently estimating cell proportions for six major cell-types from bulk mRNA expression⁴¹. Conveniently, this method is freely available for researchers (R package, BRETIGEA), which will facilitate reproducibility analyses of our study. Other important considerations are the dynamic nature of gene expression as disease progresses^{102, 103}, *post-mortem* RNA degradation of the used templates¹⁰⁴, and the subsequent limited ability of bulk RNA sequencing to reflect cell-to-cell variability, which is relevant for understanding cell heterogeneity and the roles of specific cell populations in disease¹⁰⁵. Lastly, a promising future direction would be to validate our findings with single-cell spatial analyses alongside cellular interaction modeling with pluripotent stem (iPS) cells.

Disease-specific atrophy maps

Voxel-wise brain atrophy maps in early- and late-onset Alzheimer's disease (EOAD and LOAD), Parkinson's disease (PD), amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), dementia with Lewy bodies (DLB) and dementia-related pathologies (presenilin-1, TDP43A, TDP43C, 3RTau, and 4RTau) were adopted from open data repositories and/or requested from collaborators^{32, 33}, as specified below. Reduction in gray matter (GM) density in diseased atrophy maps relative to controls was measured by voxel- and deformation-based morphometry (VBM and DBM) applied to structural T1-weighted MR images, and thus were expressed as t-score per voxel (relatively low negative t-scores indicate greater GM tissue loss/atrophy)^{36, 37}. VBM is a hypothesis-free technique for analyzing neuroimaging data that characterizes regional tissue concentration differences across the whole brain, without the need to predefine regions of interest¹⁰⁶. DBM is a similar widely used technique to identify structural changes in the brain across participants, which in addition considers anatomical differences such as shape and size of brain structures¹⁰⁷. See Table S2 for study-origin, sample size and imaging technique corresponding to each atrophy map.

MRI data for neuropathological dementias were collected from 186 individuals and averaged across participants per condition: 107 had a primary AD diagnosis (68 early-onset (<65 years at disease onset), 29 late onset (≥65 years at disease onset), 10 presenilin-1 mutation carriers), 25 with DLB, 11 with 3-repeat-tau, 17 with 4-repeat-tau, 12 TDP43A, and 14 TDP43C)³². Data were collected from multiple centres on scanners from three different manufacturers (Philips, GE, Siemens), using a variety of different imaging protocols³². Magnetic field strength varied between 1.0 T (n=15 scans), 1.5 T (n=201 scans) and 3 T (n=43 scans)³². Atrophy maps were statistically adjusted for age, sex, total intracranial volume, and MRI strength field and site³². Ethical approval for this retrospective study was obtained from the National Research Ethics Service Committee London-Southeast³².

MRI data for PD, consisted of 3T high-resolution T1-weighted scans, were obtained from the Parkinson's Progression Markers Initiative (PPMI) database (www.ppmiinfo.org/data) (<http://www.ppmiinfo.org/data>)⁸³. The PPMI is a multi-center international study with approved protocols by the local institutional review boards at all 24 sites across the US, Europe, and Australia⁸³. MRI and clinical data used in constructing atrophy maps were collected from 232 participants with PD⁸³. PD subjects (77 females; age 61.2 ± 9.1) were required to be at least 30 years old or older, untreated with PD medications within 2 years of diagnosis, and to display at least two or more PD-related motor symptoms⁸³. No significant effect of age, gender, or site was found⁸³.

For ALS, MRI data was collected from 66 patients (24 females; age 57.98 ± 10.84) with both sporadic or familial form of disease from centers of the Canadian ALS Neuroimaging Consortium (<http://calsnic.org/>)¹⁰⁸, *ClinicalTrials.gov* (<http://clinicaltrials.gov/>)¹⁰⁸ NCT02405182), which included 3T MRI sites in University of Alberta, University of Calgary, University of Toronto, and McGill University¹⁰⁸. All participants gave written informed consent, and the study was approved by the health research ethics boards at each of the participating sites¹⁰⁸. Participants were excluded if they had a history of other neurological or psychiatric disorders, prior brain injury or respiratory impairment resulting in an inability to tolerate the MRI protocol¹⁰⁸. Normative aging as well as sex differences were regressed out from data prior the map construction¹⁰⁸.

For FTD, MRI data was accessed from the frontotemporal lobar degeneration neuroimaging initiative (FTLDNI), which included 125 patients (73 females, age 63.52 ± 7.03) with frontotemporal dementia diagnosis¹⁰⁹. 3T structural images were collected on three following sites: University of California San Francisco, Mayo Clinic, and Massachusetts General Hospital¹⁰⁹. All subjects

provided informed consent and the protocol was approved by the institution review board at all sites.¹⁰⁹ For up-to-date information on participation and protocol, please visit: <http://memory.ucsf.edu/research/studies/nifd>.

Mapping gene expression data

To construct a comprehensive transcriptome atlas, we used RNA microarray gene expression data from the Allen Human Brain Atlas (AHBA; <https://human.brain-map.org/>).³⁹ The AHBA included anatomical and histological data collected from six healthy human specimens with no known neurological disease history (one female; age range 24–57 years; mean age 42.5 ± 13.38 years).³⁹ Two specimens contained data from the entire brain, whereas the remaining four included data from the left hemisphere only, with 3702 spatially distinct samples in total.³⁹ The samples were distributed across cortical, subcortical, brainstem and cerebellar regions in each brain, and quantified the expression levels of more than 20,000 genes.³⁹ mRNA data for specific brain locations were accompanied by structural MR data from each individual and were labeled with Talairach native coordinates and Montreal Neurological Institute (MNI) coordinates, which allowed us to match samples to imaging data.

Following the validated approach in⁴⁰, missing data points between samples for each MNI coordinate were interpolated using Gaussian-process regression, a widely used method for data interpolation in geostatistics. MNI coordinates for missing mRNA values were taken from the GM regions of the AAL atlas. Spatial covariance between coordinates from 3072 AHBA tissue samples and coordinates from the AAL atlas was estimated via the quadratic exponential kernel function. mRNA expression at each MNI coordinate was then predicted by multiplying AHBA gene expression values that corresponded to specific probes to kernel covariance matrix divided by the sum of kernels.

Cell-type proportion estimation

Densities for multiple canonical cell-types were estimated at the gray matter (GM) by applying an R-package Brain Cell-type Specific Gene Expression Analysis (BRETIGEA), with known genetic markers to the transcriptome atlas.⁴¹ This deconvolution method was designed for estimating cell proportions in bulk mRNA expression data for six major cell-types: neurons, astrocytes, oligodendrocytes, microglia, endothelial cells, and oligodendrocyte precursor cells.⁴¹ We chose 15 representative gene markers per each cell-type (90 in total) from the BRETIGEA human brain marker gene set and then selected those genes that were also present in the AHBA gene expression database with matching gene probes. This resulted in eighty cell-type-related gene markers that were used in missing data interpolation and the deconvolution and proportion estimation analysis (Table S3). For each voxel, each cell-type proportion value was normalized relative to the sum of all six cell-types and the sum was scaled relative to the gray matter density. We then registered data into MNI and volumetric space using the ICBM152 template.³⁸

For the correlation analysis, cell densities were averaged over 118 anatomical regions in gray matter defined by the extended automated anatomical labelling atlas (AAL; Table S1).⁴² We repeated the correlation analysis for the 98 regions from the Desikan-Killiany-Tourville atlas (DKT; Fig. S1).⁴³

Data analysis

We constructed a 6×11 correlation matrix by computing inter-regional Spearman's correlations between spatial distributions of the six canonical cell-types and patterns of atrophy in eleven neurodegenerative conditions. Shapiro-Wilk tests were used to examine the normality of data

distribution. Hierarchical clustering analyses was applied using in-built MATLAB function for data visualization. Samples were clustered together based on estimated averaged linkage Euclidian distance between them.

Data and materials availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. The BRETIGEA R package can be downloaded from <https://cran.r-project.org/package=BRETIGEA>. The Allen Human Brain Atlas data is available at <https://human.brain-map.org/static/download>. Atrophy maps for pathologically confirmed dementia are available at <http://neurovault.org/collections/ADHMHOPN/>. Raw demographic and MRI data from PD and ALS patients can be accessed at www.ppmiinfo.org/data (<http://www.ppmiinfo.org/data>) and <http://calsnic.org/> (ClinicalTrials.gov (<http://clinicaltrials.gov/>) NCT02405182), respectively. We anticipate that resulting cells abundance maps will be freely shared with the community with article publication (at our lab's GitHub space <https://github.com/neuropm-lab>).

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Competing interests

The authors declare no competing interest.

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Reviewer #1 (Public Review):

This study explores the relationship between the most common spatial patterns of neurodegeneration and the density of different cell types in the cerebral cortex. The authors present data showing that atrophy patterns in Alzheimer's disease and Frontotemporal dementia strongly associate with the abundance of astrocytes and microglia. While the results here may be considered preliminary, this work takes a step in the right direction by emphasizing the critical role that cells other than neurons play in the degeneration patterns observable with neuroimaging.

I have two main comments:

1. The authors make an important innovation by applying the cellular deconvolution approach to create brain-wide maps of cellular abundance, and then comparing these maps to atrophy patterns from the most common neurodegenerative diseases and dementia syndromes.
2. I would have preferred to see more figures with brain images showing the cellular abundance maps and the atrophy maps. Without being able to see these figures, it's difficult for the reader to 1) validate the atrophy patterns or 2) gain intuition about how the cellular abundance maps vary across the brain. The images in Figure 1C give a small preview, but I'd like to see these maps in their entirety on the brain surface or axial image slices.

Reviewer #2 (Public Review):

Pak et al. report on a study using a computational method to assess differences in the relative proportion of six canonical brain cell types, across eleven neurodegenerative classes (defined as both clinical syndromes (e.g. FTD, PD), groups of neurodegenerative diseases (e.g. 4-repeat tauopathies) or distinct neuropathological entities (e.g. FTLTDP type C), as they relate to a standard map of class-dependent volume loss. The study uses innovative methods and is commendable in its goal to highlight the contribution of non-neuronal cell types to the pathobiology of neurodegeneration. The findings of the study are in part contradicting expected results based on extensive literature on the biology of these diseases. The authors based their methodology on the use of a deconvolutional cell classifier; however, do not extensively recognize that their data on gene expression are based on normal brain levels rather than on diseased ones. Also, while predicted levels are uniquely based on patterns of brain atrophy, it is not possible to know whether this strategy is generalizable to all diseases (for instance, it is known that pure DLB, PD and ALS are not associated with extensive brain atrophy), or even adequately comparable between subtypes of diseases within the same class (e.g., different forms of FTLTDP). The authors do not acknowledge that only data based on true neuropathological assessment may prove whether their findings are true. Subject characteristics, numbers, and diagnostic criteria are hard to assess and only described in the methods section. This format prevents the reader from assessing data robustness while going through the results, especially when fundamental biological bases of nomenclature and differences between clinical syndromes and pathological entities are omitted or uncharacteristically provided.

Reviewer #3 (Public Review):

This study is a fine example of a recent productive trend in the integration of neuroimaging and molecular biology of the brain: in brief, overlaying some neuroimaging data (usually from a large cohort) onto the high spatial resolution gene expression in the Allen Human Brain Atlas data, derived from 6 individuals. By projecting structural MRI images over cell type proportions identified in the Allen data, the authors can represent various diseases in terms of their spatially-associated cell types. The result has implications for prioritizing the contributions of various cell types to each disease and creates an even-handed cell type profile through which the 11 diseases can be compared.